

## 1. Introduction

Now that you understand the science, tools, and preparation behind air layering, this module focuses on execution mastery. This is where knowledge becomes skill.

You will learn:

- A refined, professional step-by-step process
  - Advanced variations of marcotting
  - Troubleshooting real-world problems
  - Techniques for different plant species
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## 2. The Complete Professional Workflow

This is the standardized method used by experienced growers.

### Step 1: Branch Selection

- Choose a healthy, disease-free branch
- Diameter: 1–2 cm (pencil-thick)
- Type: Semi-hardwood
- Position: Upright or angled

### Step 2: Girdling the Branch

- Make two clean circular cuts (2–3 cm apart)
- Remove the bark completely

### Step 3: Cambium Removal

- Scrape the green cambium layer
- Ensure the surface feels dry, not slippery

### Step 4: Apply Rooting Hormone

- Apply to the upper cut surface
- Use powder or gel form

### Step 5: Apply Rooting Medium

- Wrap moist sphagnum moss or coconut coir
- Ensure full coverage around the wound

### Step 6: Wrap and Seal

- Cover with plastic wrap

- Seal tightly at both ends
  - Add aluminum foil (optional but recommended)
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### 3. Advanced Marcotting Techniques

#### A. Double Air Layering

- Apply two marcots on one branch
- Used for maximizing propagation output

Important: Ensure enough spacing between layers to avoid stress.

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#### B. Serpentine Layering

- Used for flexible branches
  - Multiple rooting points along a single branch
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#### C. Tourniquet Method

- Instead of removing bark, tightly wrap wire
- Restricts nutrient flow

Use Case: For delicate plants where girdling may cause damage

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#### D. Ring vs. Slit Method

| Method          | Description       | Best For     |
|-----------------|-------------------|--------------|
| Ring (Girdling) | Full bark removal | Most plants  |
| Slit Method     | Vertical cut only | Softer stems |

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### 4. Species-Specific Adjustments

Different plants require slight modifications.

Fruit Trees (e.g., Mango, Citrus)

- Require full girdling
- Benefit from rooting hormones

Ornamental Plants (Ficus, Rubber Tree)

- Root faster

- Can tolerate less precise conditions

#### Hardwood Species

- Take longer (6–10 weeks)
  - Require strict moisture control
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### 5. Monitoring Progress Like a Professional

#### Weekly Checklist

- Check moisture level
- Inspect for fungal growth
- Observe root formation

#### Signs of Success

- White roots visible through plastic
- Gradual thickening of root mass

#### Signs of Failure

- Dry medium
  - Black or mushy tissue
  - No root growth after several weeks
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### 6. Troubleshooting Common Problems

#### Problem 1: No Roots Forming

##### Causes:

- Cambium not removed properly
- Lack of moisture

##### Solution:

- Re-scrape and rewrap
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#### Problem 2: Rotting at the Site

##### Causes:

- Overwatering
- Poor sterilization

Solution:

- Remove affected section
  - Restart with clean tools
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### Problem 3: Roots Form but Die After Cutting

Causes:

- Roots harvested too early
- Poor aftercare

Solution:

- Wait for mature roots (slightly brown)
  - Improve acclimatization
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## 7. Harvesting the Marcotted Branch

When to Cut

- Roots are:
  - Thick
  - Numerous
  - Slightly woody

How to Cut

- Use sterilized pruning shears
  - Cut just below the root ball
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## 8. Transplanting Technique

Steps

1. Carefully remove plastic
2. Do NOT disturb the root ball
3. Plant in well-draining potting mix
4. Water lightly

Critical Rule

Never shake or remove the rooting medium—this protects fragile roots.

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## 9. Acclimatization (Hardening Phase)

This is where many beginners fail.

Process

- Keep in shade for 1–2 weeks
- Maintain humidity
- Gradually expose to sunlight

Goal

Help the plant transition from:

- Dependent (mother plant)  
→ Independent (own root system)

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## 10. Demonstration Scenario

Case Study: Rubber Tree Marcotting

- Week 1–2: Callus formation
- Week 3–4: Root emergence
- Week 5–6: Root thickening
- Week 6+: Ready for harvesting

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## 11. Module Summary

By the end of this module, you should be able to:

- Perform air layering with confidence
- Apply advanced techniques
- Adapt methods to different plant species
- Diagnose and fix common problems
- Successfully harvest and transplant marcotted plants